6.6.2 Double Acting Pump

Construction

This type of pump is similar to the single acting pump in its construction. However, it differs from the single acting pump that the plunger in the pump is operated in such a way to achieve the suction and delivery of fuel at the same time. Figure 6.6.2 shows the Double Acting Pump.

Working Principle

When the drive is received from the engine cam shaft, the cam shaft of the FIP starts rotating and actuates the roller tappet of the fuel pump. Hence the plunger moves downward and presses the pressure spring. Now the pressure valve and suction valve are opened at the same time. And the fuel suction and delivery are achieved at the same time. For each and every rotation of FIP camshaft, the pressurized fuel delivery and suction process are done simultaneously. Since the two actions are done at the same time, this pump is called as the double acting pump.

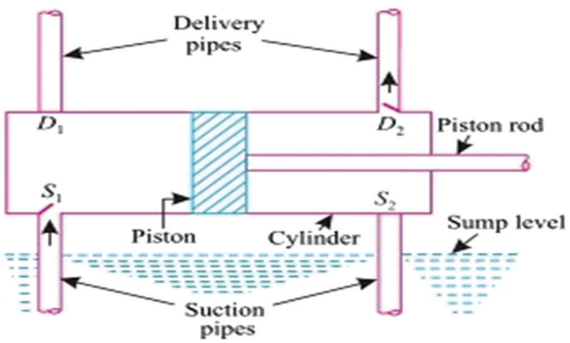


Figure 6.6.2 Double Acting Pump

6.7 Inlet Manifold

Inlet manifold is a pipe which is made up of cast iron or aluminium metal. It is used to send the air fuel mixture from the carburettor to the engine inlet port in petrol engines and air to the combustion chamber in diesel engines. It is also a place to locate the carburettor in petrol engines. The governor or super charger can't be fitted here. In L-head engine the inlet manifold is situated at the adjacent of cylinder block, and in I- head engine it is situated at the adjacent of cylinder head. In V- shape engine it is situated between the two cylinders. Since the exhaust and coolant supply are placed close to the intake manifold there may be possibilities for the inlet manifold to get heated. Hence a thermostat valve with coolant circulation is provided in modern engines to avoid heating of the intake manifold. Therefore, by using the thermostat valve, the inlet manifold can be cooled. The engine is also heated easily by using the thermostat valve from low temperature. Therefore, engine can be started easily. The inlet manifold of the engine is classified into two types. They are

1. Dual intake manifold
2. Four-barrel intake manifold



Dual Intake Manifold



Four-Barrel Intake Manifold.

Figure 6.7 Dual and Four-Barrel Intake Manifold.

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